



SECJ3744 ENTERPRISE SYSTEMS DESIGN & MODELLING

Technical Report

GROUP ASSIGNMENT (10%)

**Using AI, Blockchain, and Analytics to Promote Sustainability Reporting
and Green Supply Chains**

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EXECUTIVE SUMMARY

Companies need to provide increasingly precise and trustworthy information about sustainability performance to meet current expectations. It is a challenge that on-premises Enterprise System (ES) are usually unable to meet. This report discusses how technologies can address this challenge with the use of Artificial Intelligence (AI), Blockchain, and Data Analytics. AI relies on predictive models to make accurate demand planning and resource allocation, which reduces waste dramatically. Then, Blockchain technology establishes a digital record, the immutable ledger, which cannot be tampered with. By promoting goods traceability to avoid counterfeiting, and proof of the ethical origin, companies can achieve it. Lastly, complex information is gathered and formatted using advanced Analytics tools, such as BI dashboards, which create clean reporting information about Environmental, Social, and Governance (ESG). Together these technologies are critical for Building transparent green supply chains to meet the needs of consumers and regulators. By shifting sustainability from the expense associated with adherence to procedures to the competitive advantage, companies will be able to achieve the optimisation of frameworks with green supply chains.

1.0 INTRODUCTION

Sustainability has today become one of the core objections in business and enterprise systems. Enterprises are now not only focused on gross margins but also charge time and ensure corporate social responsibility (CSR). However, neither traditional Enterprise Systems (ES) nor Enterprise Resource Planning (ERP) systems were purpose-built to do so. With a lack of interoperability, arduous audit processes, and manual data collection, both make the issue of sustainability reporting tedious, costly, and susceptible to human error.

To solve this critical issue, companies have begun integrating a combination of three innovative technologies, which are AI, Blockchain, and Analytics. The method cuts directly waste and carbon emissions and gives the ability to optimise resource allocation and improve demand forecasting. Blockchain offers an immutable ledger, which provides unmatched traceability and trust for materials. It results in the reduction of fraud, and the capacity to verify universal good practices in the green supply chains. Finally, Analytics and Business Intelligence (BI) dashboards turn raw input into unambiguous and instantaneous indicators. It automates the whole reporting process. (Zhao, 2024)

The primary objective of this technical report is to analyse how these three technologies function together within an enterprise framework to promote sustainability. This paper will first provide a background on the current challenges in sustainability reporting. It will then analyse the specific role of AI, Blockchain, and Analytics, discussing their advantages, disadvantages, and implementation challenges. Finally, this report will provide practical recommendations for companies seeking to adopt this integrated approach to build more transparent, efficient, and credible green supply chains.

2.0 BACKGROUND / LITERATURE REVIEW

Companies are now required to provide strong Environmental, Social, and Governance (ESG) reports. This new pressure has shown that older enterprise systems have big weaknesses. For a long time, researchers have studied Green Supply Chain Management (GSCM). The primary purpose of green supply chain management (GSCM) is to reduce the ecological footprints of an organization along the business process chain, from carrier procurement to product delivery. However, research consistently identifies a persistent issue, which is poor data and visibility of the supply chain in businesses. Researchers said GSCM initiatives frequently fail when there is no reliable method to identify and verify demands for sustainability (Cherono et al., 2025). Sustainability reports become difficult to validate as a result.

Currently, innovations in technology are being examined to overcome the challenge. The majority of the work will be centred around utilising artificial intelligence (AI) to enhance supply chain processes. For example, recent research indicates that AI-based demand forecasting models provide a means to minimize overproduction and eliminate waste (Kalpana et al., 2025). Likewise, firms make use of predictive analytics to identify optimal trucking routes, which results in fuel savings and a reduction in carbon emissions.

As well, blockchain technology is being explored as a means to increase transparency among different suppliers along the supply chain. The singular most unique characteristic that the blockchain system boasts is its permanent ledger, which is actually a computerized immutable recording mechanism. This is normally seen as the best solution for creating a single trustworthy system of traceability. (Doshi et al., 2024). This is particularly useful when it comes to authenticating and verifying the original material source or whether an organization adheres to environmental policies. Researchers similarly find that BI dashboards are the other major trend. It allows managers to assess live ESG information using straightforward charts (Khaddam & Alzghoul, 2025).

However, all these technologies are investigated in isolation in the majority of works. Very little information is available regarding the integration of all three technologies, AI, Blockchain, and Analytics. The purpose of this report is to analyse how a combination of these tools forms a system that not only creates efficient supply chain and sustainability reporting system.

3.0 ANALYSIS AND DISCUSSION

3.1 AI, BLOCKCHAIN, ANALYTICS IN SUSTAINABILITY

Technology	Core Function	Contributions to Sustainability
Artificial Intelligence (AI)	Uses predictive models and machine learning to find patterns, forecast, and optimise	Reduces waste by accurately predicting future demand. Cuts emissions by finding the most fuel-efficient truck routes.
Blockchain	A shared digital record book that is immutable ledger. It uses smart contracts to automate agreements.	Proves origins by creating a trusted trail for all materials. Prevents fraud and verifies claims. Builds trust by making reports auditable.
Analytics	Tools that clean, process, and visualise data, including predictive analytics and BI dashboards.	Creates clear reports by turning raw data into simple charts. Tracks performance by showing live ESG data. Helps managers make decisions to improve sustainability.

3.2 REAL-WORLD APPLICATIONS

These technologies can be seen even in current business applications. For example, Maersk is a global logistics giant that employs blockchain and machine learning to monitor cargo and cut down inefficiency (Wong et al., 2023). In farming, AI algorithms can forecast optimal planting schedules and deployment of resources, while blockchain follows produce along the journey from farmland to supermarkets to guarantee that it was harvested ethically. In manufacturing, AI ensures manufacturing is optimised to cut down wastage, whilst analytics software approximates the carbon footprint in factories.

3.3 ADVANTAGES & BENEFITS

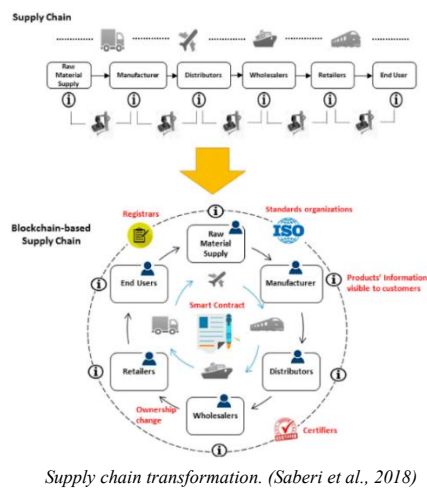
These technologies offer several benefits. First, they build transparency and confidence through guaranteeing that sustainability data are always trackable and protected. Second, these technologies enhance efficiency through automating tracking as well as reporting of environment-related facts with minimal human error and administrative complexity. Thirdly, these technologies enable better decision making since trends and relationships are revealed

through analytics which managers can then apply to uplift sustainability performance. Finally, these technologies enable compliance with regulatory requirements as well as reputation enhancement among customers, investors, as well as governments (Dubey et al., 2017).

3.4 CHALLENGES & RISKS

There are still several big challenges to overcome. First, these systems are very expensive, and the high cost of installing sensors, cloud platforms, and software is often too much for small or medium-sized businesses (Tijan et al., 2019). Second, data privacy are major issues. Companies must find ways to protect sensitive supplier and employee information, especially when recording it on a permanent blockchain ledger (Mohammed et al., 2024). An important concern about interoperability is that it is not easily achieved when companies use different ERPs, which would hinder the interaction between the various systems involved. Blockchain technology has its weaknesses as well, for example, scalability issues that make certain supply chains slow-moving due to their large size, and how certain systems use a great deal of energy, which is not environmentally friendly. The last limitation is human factors that can impede progress, including employees' lack of digital literacy, companies being slow to update, and suppliers opposing the new platform (Ghode et al., 2021).

3.5 FUTURE TRENDS



One of the new trends in real life is using AI tools like Large Language Models (LLMs) to help write sustainability and ESG reports. These tools can read long reports, find the key information, and write summaries. Thus, it will save time and keeps the information clear and accurate. For example, *ESGReveal* already uses AI to make reporting easier and to help companies follow government rules. Another trend is to digitalise carbon credits by applying blockchain. Each digital token represents the value of one real carbon credit (Segaran et al., 2025). These digital tokens can subsequently be verified by artificial intelligence to confirm they are real. This makes the companies transparent, honest, and green.

4.0 RECOMMENDATIONS

To use the most value from AI, blockchain, and analytics for sustainability, businesses should have a clear and realistic agenda aligned to their long-term environmental ambition. Pilots are

preferable to making wholesale transformation all at once. Businesses, for example, can begin by piloting smaller projects like tracking suppliers or tracking carbon emissions. They are then able to pilot the system, obtain the costs, and build the confidence of the personnel involved prior to scaling to the whole organisation.

Second, companies should bring together AI, blockchains, and analytics in the same system so data moves effortlessly between systems. data management needs a strong set of rules, which defines who owns the data, how the data is kept confidential, and how the authenticity of the data is assured. By adopting GRI, SASB, and TCFD, the sustainability reports become simpler to compare and more comprehensible.

Moreover, employees, suppliers, and management need to be trained on digital tools and sustainability processes, so they can fully utilize the technology. Companies should also cooperate with regulators, auditors, and technology providers, so systems conform with legal frameworks and codes of conduct. And, of course, disclosure is critical. The companies should be transparent and open; third parties should be hired to audit sustainability reports.

5.0 CONCLUSION

This paper showed how the use of AI, Blockchain, and Analytics in combination can resolve many of the challenges of sustainability reporting. These technologies make it possible to increase green supply chain trustworthiness, efficiency, and transparency. AI allows to forecast demand and utilize resources better, helping to reduce waste for which it is not too late to reuse. Blockchain offers an immutable digital ledger that makes it easy to prove the supply chain of materials and secure the trust of customers and business partners. Finally, Analytics turns all of this validated information into clean, real-time ESG dashboards that enable quicker and more reliable sustainability reporting.

The takeaway is that those companies that use these technologies are not just complying with regulations. They are gaining a competitive edge in the market. Their operations become more efficient, transparent, and trustworthy for both consumers and investors. Future research should investigate a global blockchain-based ESG data standard and how to integrate blockchain with IoT devices for automatic data capture. This will enable the creation of more intelligent, sustainable enterprises.

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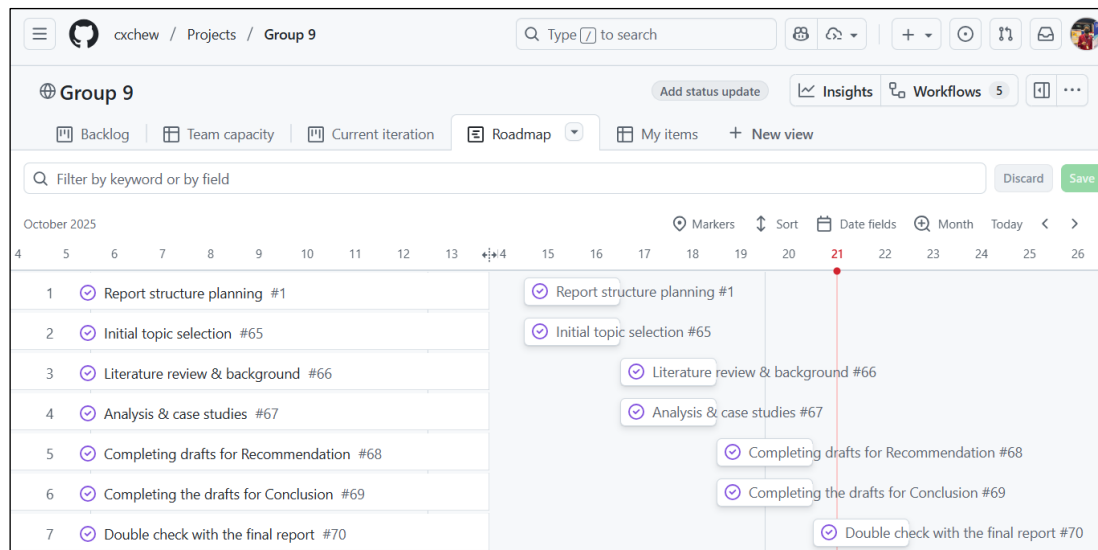
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APPENDIX

I. Logbook



Date	Progress Update	Keywords Used	References Found
15 Oct	Initial topic selection and report structure planning. Defined our focus: how AI, Blockchain, and Analytics work together in Enterprise System to improve sustainability reporting and green supply chains.	"AI sustainability supply chain", "Blockchain green logistics", "ESG data analytics"	Found an IEEE journal article on using blockchain for material traceability.
17 Oct	Literature review & background. Focused on showing why current sustainability report is difficult, and how these new technologies solve them.	"Challenges in sustainability reporting", "ERP sustainability module", "Predictive analytics for carbon footprint"	Found a report detailing specific case studies of companies using analytics dashboards for ESG metrics.
18 Oct	Analysis & case studies. Focused on finding examples for the 'Analysis and Discussion' section. Specific applications of the technologies needed.	"Robotic Process Automation (RPA) in supply chain reporting", "Cloud-based ERP sustainability features"	Found a detailed case study about a global company using AI-powered analytics to optimize transportation routes.
21 Oct	Completed the drafts for Recommendation and Conclusion sections. Double checked the document followed the guideline provided.	"Strategic recommendations sustainability tech", "Future of Work with Enterprise IS"	Confirmed all references were credible.

II. *Record of Generative AI Tools Used*

Date	AI Tool Used	Prompts Given	Purposes	Verification
17 Oct	ChatGPT	Give me 3 real-world case study examples of companies using blockchain for supply chain transparency.	To get examples for Real-world application part	AI suggested Maersk, optimal planting and carbon footprint estimation in factories. We then used Google Scholar to find the actual academic articles and journals about these information to ensure the information is credible.
17 Oct	ChatGPT	What is the difference between interoperability and scalability in blockchain and ERP systems	To understand the technical terms in blockchain for Challenge part	AI provided clear definition. We verified these definition against the technical articles to confirm there were accurate.
18 Oct	Gemini	What are the main challenges and risks of implementing AI and Blockchain in enterprise systems for sustainability?	To understand the challenge and complete section 3.	AI listed high cost, interoperability, and scalability. We confirmed these were major themes by cross-referencing them with academic sources.
18 Oct	Gemini	Give me 3-4 key future trends for AI and blockchain in green supply chains	To have a rough idea for future trends part	AI suggested Generative AI, Carbon Tokenisation, and IoT integration. We used these ideas as a starting point for our own research, which led us to find our sources on LLMs and tokenisation.
19 Oct	Gemini	Provide 5 practical recommendations for a company that wants to use AI and blockchain for sustainability reporting	To have some ideas for recommendation part	We agreed the AI suggestions and used them as the basis for our recommendations where we then do research on that.
21 Oct	Gemini	Crosscheck my report with the rubric given.	To make sure our report is aligned with the rubric.	AI crosscheck the report with rubric given to ensure the report is aligned with the project instruction.